

EE 5239 Nonlinear Programming

Lecture 13a: Sparse and Low Rank Optimization

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Announcement

- Plan for the next three lectures
 - (1) Optimization methods for sparse and low-rank problems
 - (2) Overview of zeroth-order optimization with application to adversary ML

This Time

- Next two lectures, **sparse/low-rank** optimization and modeling
- A.k.a large-scale “structured optimization”
- Modern optimization paradigm
- Active research areas
- Again focus on applications, algorithms, and theories

Application 1: Linear Regression

- 1 **Training data sets** ($\mathbf{a}_i$ is the feature, and b_i is the label)

$$\{\mathbf{a}_i, b_i\}_{i=1, \dots, M}, \mathbf{a}_i \in \mathbb{R}^K, b_i \in \mathbb{R}$$

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$$f(\mathbf{x}; \mathbf{a}) = \mathbf{a}^T \mathbf{x}$$

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- ③ We have absorbed x_0 in $\mathbf{x}$ and augmenting $\mathbf{a}_i$'s with extra 1
④ Use the cost function

$$L(\mathbf{x}) = \underbrace{\frac{1}{2} \sum_i (\mathbf{a}_i^T \mathbf{x} - b_i)^2}_{\text{squared loss}} = \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

Minimize w.r.t. $\mathbf{x}$: $\mathbf{x}^* = \left(\mathbf{A}^T \mathbf{A}\right)^{-1} \mathbf{A}^T \mathbf{b}$ (If $\mathbf{A}^T \mathbf{A}$ invertible)

Application 1: Linear Regression

- In practice, usually look for **simple, approximate** solutions of regression (or for more general learning problems)
 - ① **Occam's Razor**: Simple explanations of the observations are preferable to complex explanations
 - ② Care about **a few** key characteristics that are most important
 - ③ In certain problems the dimension of the variable is huge (x very long) while the data is scarce (M is small) (i.e., $\mathbf{A}$ is a very “fat” matrix)
 - ④ Simple solutions are often **easy to explain/implement/store**
 - ⑤ Conform to **prior knowledge**
- In regression, what does a “**simple solution**” mean?

Application 1: Linear Regression+Parameter Selection

- ① In certain applications (e.g., bioinformatics), getting data samples is expensive
- ② But the number of variables/features are huge
- ③ This means that the data matrix $\mathbf{A}$ is “fat”
- ④ If all features are used, the linear equation $\mathbf{A}\mathbf{x} = \mathbf{b}$ is under-determined
- ⑤ Lots of solutions! But which are good solutions?

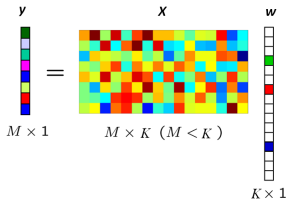
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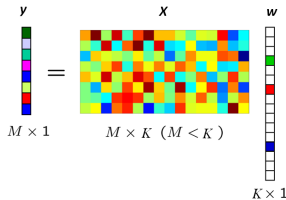
$$\{\mathbf{a}_i, b_i\}_{i=1, \dots, M}, \mathbf{a}_i \in \mathbb{R}^K, b_i \in \mathbb{R}$$

- ② Objective: Learn a predictive function,

$$f(\mathbf{x}; \mathbf{a}) = \mathbf{a}^T \mathbf{x}$$

- ③ Use the cost function (**Add a regularizer!**)

$$L(\mathbf{x}) = \underbrace{\frac{1}{2} \sum_i (\mathbf{a}_i^T \mathbf{x} - b_i)^2}_{\text{squared loss}} + \underbrace{R(\mathbf{x})}_{\text{regularizer}}$$



Application 2: LR/SVM+ Parameter Selection

- Similarly, we can add regularizers to the objective of LR/SVM
- Regularized LR

$$L(\mathbf{x}) = \frac{1}{M} \sum_{i=1}^M \ln \left(1 + e^{-b_i \mathbf{a}_i^T \mathbf{x}} \right) + R(\mathbf{x})$$

- Regularized SVM

$$L(\mathbf{x}) = \frac{1}{M} \sum_{i=1}^M \max\{0, 1 - b_i \mathbf{a}_i^T \mathbf{x}\} + R(\mathbf{x})$$

Application 3: Image Processing

- **Image Denoising:** Given a matrix of “noisy” pixel intensities $\mathbf{Y}$, find a “clean” matrix $\mathbf{X}$
- **Requirement:**
 - ① $\mathbf{X}$ close to $\mathbf{Y}$
 - ② $\mathbf{X}$ contains less noise, i.e., “smoother”

Application 3: Image Processing

- **Key Idea** Adjacent pixels look “alike”

$$\min_{\mathbf{X}} \underbrace{\|\mathbf{X} - \mathbf{Y}\|^2}_{\text{Data Fidelity}} + \underbrace{R(\mathbf{X})}_{\text{Regularization}}$$

- What's the requirement for $R(\mathbf{X})$ here?

Application 3: Image Processing



Figure 0.1: Original v.s. Noisy Images [S. Wright 2009]



Figure 0.2: Denoised Images [S. Wright 2009]

Application 4: Compressive Sensing

- In a MRI system, we have measurement in the form of

$$\mathbf{b} = \mathbf{A}\mathbf{x} + \mathbf{n}$$

where $\mathbf{x} \in \mathbb{R}^K$ is the image of interest (vectorized); $\mathbf{n} \in \mathbb{R}^M$ is measurement noise, and $\mathbf{A} \in \mathbb{R}^{M \times K}$ is the measurement matrix (partial Fourier Transform Matrix)

- Large **under-determined** linear system $M \ll K$, to reduce data volume, or the number of sensed samples



Figure 0.3: MRI, Sampling Points, Image

Application 4: Compressive Sensing

- We have an underdetermined and noisy **recovery** problem
- What we can utilize is the fact that we know x is **sparse**
- Different formulations

$$\min R(x), \quad \text{subject to } Ax = b,$$

$$\min \|Ax - b\|^2 + R(x)$$

$$\min \|Ax - b\|^2, \quad \text{subject to } R(x) \leq s$$

- If regularizers are chosen properly these problems are convex
- If matrix A satisfies certain conditions, solving these problems exactly recovers x

Application 4: Compressive Sensing

- Pick 25% coefficients at random (with bias)
- Reconstruct image from the 25% coefficients

Application 4: Compressive Sensing

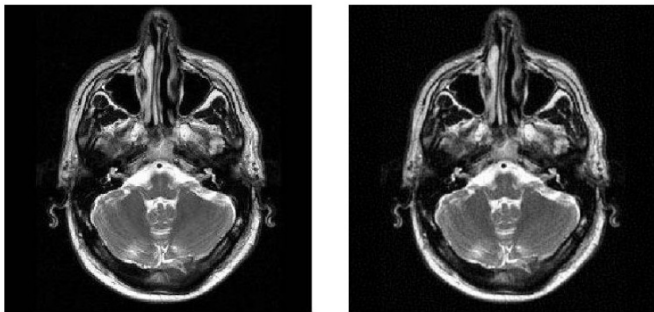


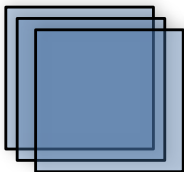
Figure 0.4: Recovery using 1/4 Fourier Coefficients: Original v.s. Reconstructed Image [Images from Y. Zhang 08]

Application 5: Anomaly Detection

- Find unusual/rare events from a large amount of data
- Lots of applications, data analysis, imaging, cybersecurity
- Utilizing the sparsity of the “anomaly”

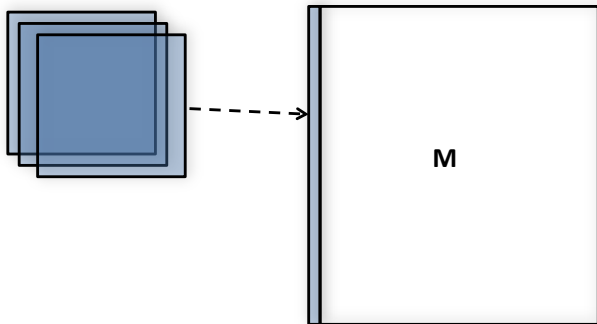
Application 5: Anomaly Detection

- An application of this idea on foreground separation from noisy surveillance video
- Suppose we have a sequence of video frames $\{\mathbf{M}^i\}_{i=1}^n$
- First we do a preprocessing:



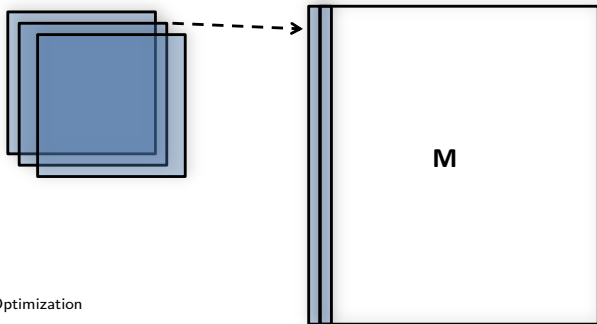
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Application 5: Anomaly Detection

- We make each column of $\mathbf{M}$ becomes **one frame** of video
- **Objective:** Decompose it into three parts:
 - ① $\mathbf{L}$ is the background (**low rank**)
 - ② $\mathbf{S}$ is the moving foreground (**sparse**)
 - ③ $\mathbf{Z}$ is the noise (**small magnitude**)
- Why low rank matrix represents background? Changes slowly; consider a matrix with identical columns, the rank is **1**
- **Note:** No “label”, unsupervised learning!
- **Main Idea:** Promote these properties by penalizing using different costs: $(\min \text{Rank}(\mathbf{L}) + \text{Density}(\mathbf{S}) + \text{Magnitude}(\mathbf{Z}))$

Application 5: Anomaly Detection

$$\begin{array}{ll} \min & \|\mathbf{L}\|_* + \gamma_1 \|\mathbf{S}\|_1 + \gamma_2 \|\mathbf{Z}\|_F \\ \text{subject to} & \mathbf{L} + \mathbf{S} + \mathbf{Z} = \mathbf{M} \end{array}$$

- 250×135 pixels per frame, 50 frames, over 1 million variables!!
- Similar idea applies to
 - 1 Analyzing power network security (intrusion detection)
 - 2 Data cleansing (getting rid of the data outliers)
 - 3 Network data flow with intrusion detection

Application 5: Anomaly Detection

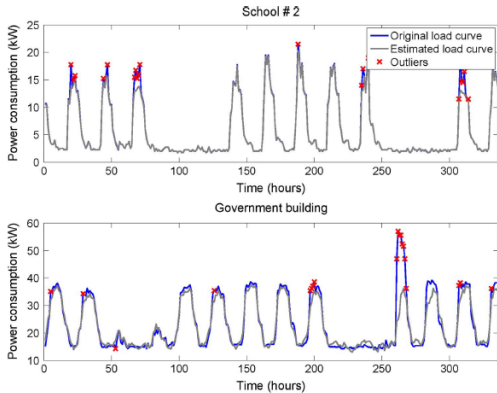


Figure 0.5: An Example in Power Data Cleansing and Imputation [Mateos and Giannakis 13]

Outline

- 1 Introduction
- 2 Applications
- 3 Formal Formulation

The Utility of Nonsmooth Norms

- Most of the problems we have seen so far can be formulated as

$$\min f(\mathbf{x}) := \underbrace{L(\mathbf{x})}_{\text{data fidelity term}} + \underbrace{R(\mathbf{x})}_{\text{regularization}}$$

- $R(\mathbf{x})$ imposes some **structure** to the solution

Enforcing Sparsity?

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- Question:** What kind of regularizer induces “sparsity” to solution?

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- Suppose we would like $\mathbf{x}$ to **approximately** minimize L , and has **as many zeros as possible**
- Question:** What kind of regularizer induces “sparsity” to solution?
- One popular candidate is the ℓ_1 norm (where $\lambda > 0$ is some given parameter)

$$R(\mathbf{x}) = \lambda \|\mathbf{x}\|_1 = \gamma \sum_{k=1}^K |x_k|$$

Why ℓ_1 norm?

- Consider the following scalar optimization problem

$$\min f(x) := L(x) + \lambda|x|$$

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$$\partial f(x) = \begin{cases} L'(x) - \lambda & \text{if } x < 0 \\ L'(0) + \lambda\epsilon & \text{if } x = 0 \\ L'(x) + \lambda & \text{if } x > 0 \end{cases} \quad (0.1)$$

where $\epsilon \in [-1, 1]$ (recall the subgradient for $|x|$)

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- Can we make this intuition precise?

Why ℓ_1 norm?

- Consider the following scalar optimization problem

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where $\epsilon \in [-1, 1]$

- The solution is (interpretation? What happens when λ is large?)

$$x^* = \begin{cases} b + \lambda & \text{if } b + \lambda < 0 \\ 0 & \text{if } |b| \leq \lambda \\ b - \lambda & \text{if } b - \lambda > 0 \end{cases} \quad (0.3)$$

Why ℓ_1 norm

- The optimal solution for problem (P)
- The solution becomes the so-called **Soft-thresholding** operation

$$x^* = \begin{cases} b + \lambda & \text{if } b + \lambda < 0 \\ 0 & \text{if } |b| \leq \lambda \\ b - \lambda & \text{if } b - \lambda > 0 \end{cases} \quad (0.4)$$

- Define $x^* = \text{Soft}(b, \lambda)$; graphically...

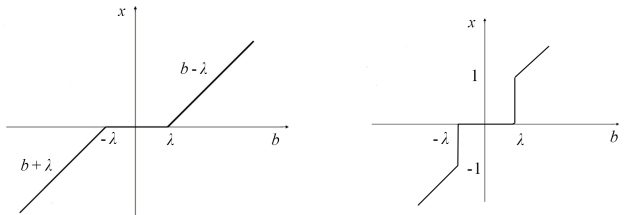


Figure 0.1: Left: The soft thresholding operation. Right: The hard thresholding operation.

Why ℓ_1 norm

- Consider the problems (enforcing exact data fidelity $Ax = b$)

$$\min \|x\|_1 \quad \text{subject to } Ax = b \quad (0.5)$$

$$\min \|x\|_2 \quad \text{subject to } Ax = b \quad (0.6)$$

$$\min \|x\|_0 \quad \text{subject to } Ax = b \quad (0.7)$$

- $\|x\|_0$ the **maximum number of nonzero components** (note: this is not a continuous function or convex function)
- $\|x\|_2$ is defined as $\sqrt{\sum_{k=1}^K x_k^2}$
- Finding the vector with the **smallest** ℓ_k norm that satisfies the **underdetermined** systems of equation $Ax = b$
- The solutions of the **intersection** of a **linear system** + ℓ_k ball

$$Ax = b, \quad \|x\|_p \leq 1, \quad p = 0, 1, 2$$

Why ℓ_1 norm

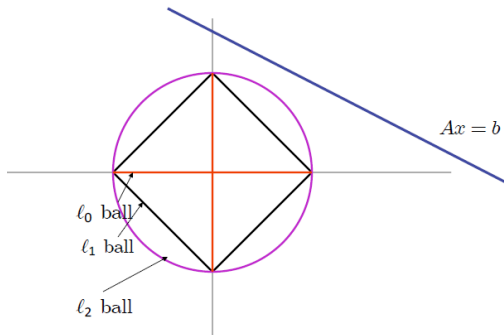


Figure 0.2: Illustration of the solution.

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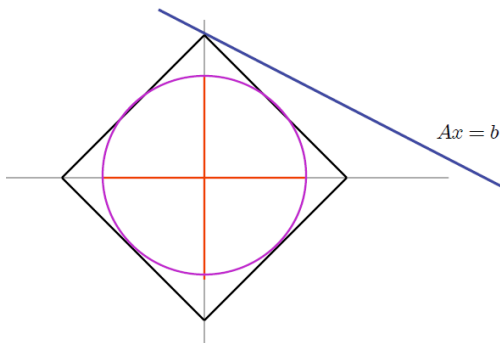


Figure 0.3: Illustration of the solution.

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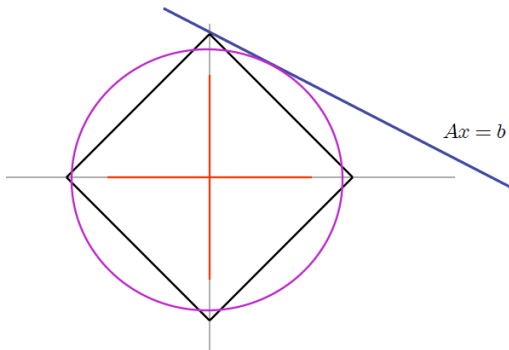


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More on Sparse Inducing Norms

- ℓ_1 norm $\|\mathbf{x}\|_1$ is a good “convex approximation” of the ℓ_0 norm (counting the number of nonzeros)
- When a single component is 0, introduces a “kink” – induces sparsity
- Other related (sparse) structure-inducing norms
 - 1 $R(\mathbf{x}) = \|\mathbf{x}\|_1$, tend to produce $\mathbf{x}$ vector with a few nonzeros
 - 2 $R(\mathbf{x}) = \|\mathbf{x}\|_2$, tend to produce $\mathbf{x}$ vector with all zeros (only kink $\mathbf{x} = 0$)
 - 3 $R(\mathbf{x}) = \sum_{I=1}^G \|\mathbf{x}_{[I]}\|_2$, where $\mathbf{x}_{[I]}$ is a group of variables; tend to produce $\mathbf{x}$ vector with a few groups of nonzeros
 - 4 $R(\mathbf{x}) = \sum_{k=1}^{K-1} |x_k - x_{k+1}|$, tend to produce $\mathbf{x}$ vector with a few nonzeros adjacent differences

Example

- Consider $\mathbf{x} \in \mathbb{R}^{1000}$, $\text{card} = 30$
- $M = 200$ noisy measurements: $\mathbf{b} = \mathbf{A}\mathbf{x} + \mathbf{v}$, $\mathbf{v} \sim \mathcal{N}(0, \sigma^2 I)$
- $\mathbf{A}_{ij} \sim \mathcal{N}(0, 1)$
- ℓ_1 reconstruction: Minimizes $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \gamma\|\mathbf{x}\|_1$

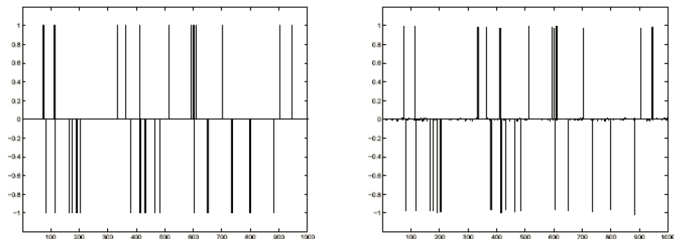


Figure 0.5: Illustration of the ℓ_1 min solution. Left: original; Right: ℓ_1 reconstruction with $\lambda = 10^{-3}$ [K. Scheinberg Lecture Notes 2012]

Example

- ℓ_2 reconstruction: Minimizes $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \gamma\|\mathbf{x}\|_2$

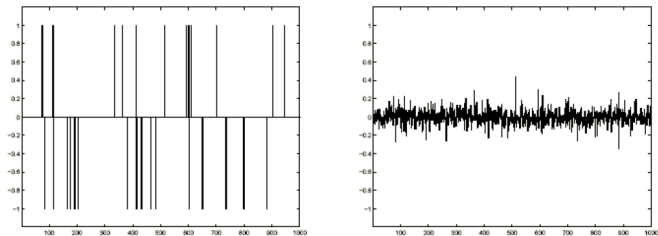


Figure 0.6: Illustration of the ℓ_1 min solution. Left: original; Right: ℓ_2 reconstruction with $\lambda = 10^{-3}$ [K. Scheinberg Lecture Notes 2012]

Relationship Between Different Formulations

- We see a few formulations to induce sparsity

Relationship Between Different Formulations

- We see a few formulations to induce sparsity
- What are their relationships?

$$\min \| \mathbf{Ax} - b \|_2, \text{ s.t. } \| \mathbf{x} \|_1 \leq s \quad \Leftrightarrow$$

$$\min \| \mathbf{Ax} - b \|_2^2, \text{ s.t. } \| \mathbf{x} \|_1 \leq s$$

$$\min \| \mathbf{Ax} - b \|_2 + \lambda \| \mathbf{x} \|_1 \quad \Leftrightarrow \quad \min \| \mathbf{Ax} - b \|_2^2 + \lambda \| \mathbf{x} \|_1$$

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Relationship Between Different Formulations

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- What are their relationships?

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$$\min \| \mathbf{Ax} - \mathbf{b} \|_2 + \lambda \|\mathbf{x}\|_1 \quad \leftrightarrow \quad \min \| \mathbf{Ax} - \mathbf{b} \|_2^2 + \lambda \|\mathbf{x}\|_1$$

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- Here, the equivalence means that
 - 1 For any s there exists a λ so that the optimal solutions are the same;
 - 2 λ corresponds to the **lagrangian multiplier** of the second constrained problem

Example: LASSO

- Consider the last pair of formulation (with parameter s)
- Make predictions on certain types of cancer
- The path of the parameters when increasing s

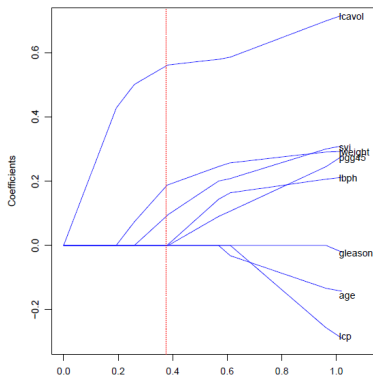


Figure 0.7: The Parameter Path for LASSO when increasing s [R. Tibshirani 2011]